Q20: LLM Internals vs. the Rest of Deep Learning

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The question

- "How much of an LLM's internals are shared with other deep learning models?"
- Why it matters: decides what knowledge transfers
- Answer preview: **the skeleton is shared; the mixer, objective, and interface are new**

The shared skeleton (everything deep)

1. Differentiable parametric function: layers of linear + nonlinearity
2. A scalar loss
3. SGD + backpropagation (reverse-mode autodiff)
4. Tensor hardware: it is all matmuls

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$\Rightarrow$ loss curves, LR schedules, init, norm, regularization intuitions all carry over

Component lineage

Component	**Origin**
SGD + backprop	1980s–90s
Embeddings	word2vec (2013)
Attention	seq2seq (2014–15)
Residual stream	ResNet (2015)
Parallel mixing (no recurrence)	**Transformer (2017)**
Next-token loss at scale	**LLM era**

Four real differences

- **Parameters:** dense huge matrices (+ MoE sparsity) vs. small shared filters
- **Loss:** self-supervised next-token vs. labeled task objectives
- **Interface:** autoregressive generation loop vs. single forward pass
- **Post-training:** SFT $\rightarrow$ preference opt (DPO/RLHF) $\rightarrow$ RLVR stack

Transfer mistakes to avoid

- Attention weights $\neq$ explanations (vision feature maps mislead here)
- Vision-style fine-tuning intuition $\neq$ prompting / ICL / LoRA economics
- "It's just a big CNN" $\neq$ correct either: the mixer and objective matter

The one-sentence map

Shared skeleton (modules + SGD + backprop + GPUs); new mixer (attention); new objective (next-token at scale); new interface (autoregression); new post-training stack.

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Shared skeleton (modules + SGD + backprop + GPUs); new mixer (attention); new objective (next-token at scale); new interface (autoregression); new post-training stack. When a new architecture appears (Mamba, diffusion LMs): check the map — the skeleton is the stable part of the field.

Reading

- Goodfellow et al., *Deep Learning*, chs. 6–8 (the skeleton)
- Vaswani et al. 2017 (the mixer change)
- Ouyang et al. 2022; Rafailov et al. 2023 (the post-training stack)